Power equations

1 Solve for the pronumeral for each of the following equations:

a $30^n = \sqrt[3]{30}$ **b** $3^y = \dfrac{1}{27}$ **c** $4^y = 64$ **d** $7^x = 1$

e $8^x = \dfrac{1}{8^2}$ **f** $\left(\sqrt{2}\right)^k = 0.5$ **g** $\dfrac{1}{36} = 6^n$ **h** $\left(\sqrt{6}\right)^y = 36$

i $5^x = \sqrt[3]{5}$ **j** $8^x = \sqrt{8}$ **k** $3^x = \sqrt[8]{3}$ **l** $4^x = 4^8$

m $x^5 = 3^5$ **n** $9^x = 9^{-4}$ **o** $3^x = 3^{\frac{2}{9}}$

2 Solve for x for each of the following equations:

a $x^{-7} = \dfrac{1}{6^7}$ **b** $7\left(4^x\right) = \dfrac{7}{4^3}$

c $10^x = 0.01$ **d** $5^{10x+33} = 125$

e $5^{-3x-1} = 3125$ **f** $\left(2^2\right)^{x+7} = 2^3$

g $\left(2^4\right)^{2x-10} = 2^2$ **h** $27\left(2^x\right) = 6^x$

i $9^y = 27$ **j** $3^{5x-10} = 1$

k $\dfrac{1}{3^{x-3}} = \sqrt[3]{9}$ **l** $30 \times 2^{x-6} = 15$

m $a^{x+1} = a^3\sqrt{a}$ **n** $25^{x+1} = 125^{3x-4}$

o $\left(\dfrac{1}{8}\right)^{x-3} = 16^{4x-3}$ **p** $3^{x^2-3x} = 81$

q $\left(\dfrac{8}{3}\right)^x = \left(\dfrac{8}{3}\right)^7$ **r** $9^{x+3} = 27^x$

Polynomial equations

3 Solve for x for each of the following cubic equations:

a $x^3 - 125 = 0$ **b** $3x^3 = 5x^2$

c $x^3 = -8$ **d** $8x^3 - 125 = 0$

e $(x+8)(x+4)(1+x) = 0$ **f** $(5x-4)(x+3)(x-2) = 0$

g $x^3 - 49x = 0$ **h** $512x^3 - 125 = 0$

i $x^3 - 3x^2 - 18x + 40 = 0$ **j** $x^3 - 4x^2 - 45x = 0$

k $x^3 + 9x^2 + 27x + 27 = 0$ **l** $-64x^3 + 48x^2 - 12x + 1 = 0$

m $x^3 - 5x^2 - 49x + 245 = 0$ **n** $x^3 + 13x^2 + 47x + 35 = 0$

o $150x^3 + 115x^2 - 118x - 56 = 0$ **p** $729x^3 + 8 = 0$

q $x^3 - 5x^2 - 4x + 20 = 0$ **r** $x(x-5)(x+7) = 8(x-5)(x+7)$

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4 Use technology to solve the following equations:

a $147x^3 + 427x^2 + 160x - 84 = 0$

b $50x^3 + 155x^2 + 152x + 48 = 0$

c $64x^3 + 40x^2 + 54x + 18 = 0$

d $48x^3 - 212x^2 + 84x + 209 = -36$

e $-576x^3 + 272x^2 - 41x + 78 = 76$

f $-25x^3 + 50x^2 - 19x - 27 = -45$

g $2x^4 + 5x^3 - 14x^2 - 5x + 12 = 0$

h $6x^4 + 9x^3 - 186x^2 + 315 = 0$

i $6x^5 - 5x^4 - 43x^3 + 70x^2 - 24x = 0$

j $5x^5 + x^4 - 25x^3 - 5x^2 + 20x + 4 = 0$

5 Solve for the roots of the cubic $P(x) = x^3 - 2x^2 - 5x + 6$ that has a factor of $x - 3$.

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